

Laptop Setup & Pre-Class Guide

Hands-On: Engineering Agentic AI Solutions — From Prototypes to Production

Welcome! This full-day, hands-on workshop moves beyond the hype to teach the core architectures — reasoning, planning, and memory — behind dependable agentic workflows. About 60% of our 6.5 hours is hands-on, so we need to start on time. **Please complete the short checklist below before Monday morning;** class time is limited, so the more you can set up in advance, the more we build together.

Critical note on corporate laptops

Enterprise laptops with restricted admin rights, local firewalls, or active VPNs frequently block Langflow Desktop or the cloud ports Google Colab uses. To avoid day-of blockers, we strongly recommend bringing a personal laptop so you have full access to every lab.

This guide has two parts. First, a quick **checklist** of what to do before class. Then a **detailed walkthrough** for installing Langflow — the one step with a few moving parts — with screenshots. If you're comfortable, the checklist alone may be enough; the walkthrough is there if you want it.

1 • Hardware & system

Requirement	Minimum	Recommended
Operating System	Windows 10/11, macOS, or modern Linux	macOS Sonoma, or Windows 11 (WSL2 preferred)
Memory (RAM)	8 GB	16 GB or higher
Local Storage	5 GB free	10 GB+ SSD for logs and dependencies
Network	Reliable internet	Stable broadband, 5 Mbps+ for Colab and APIs
Browser	Chrome, Edge, or Safari	Google Chrome v120+ (cookies enabled for Colab)

2 • Before class: three steps

Everything runs in your browser except one desktop app. There is no local Python to install and no code editor to set up — Google Colab is your editor for Labs 2–4.

Step 1 — Install Langflow Desktop

Our first lab uses Langflow, a free visual tool for building agents. Download the **Desktop app** and keep it on version **1.11.x (1.11.0 preferred)**. Do not accept the in-app update prompt — staying on 1.11.0 keeps everyone's screens matched and the lab file loading cleanly. **Full step-by-step instructions with screenshots are in Part 3 of this document.**

Download page: <https://www.langflow.org/desktop>

Step 2 — Set up Google Colab

Labs 2, 3, and 4 run in **Google Colab**, which runs in your browser — no installation, no Python setup. Just confirm you can sign in:

- Go to <https://colab.research.google.com>.
- Sign in with any Google account (a personal account is fine, and often works better than a locked corporate one).

- If you see the Colab welcome notebook, you're ready. That's the whole step.

Step 3 — Get the lab repository

All lab notebooks, data, and starter files live in one GitHub repository. You don't need to set anything up in advance — each lab notebook pulls what it needs automatically — but it helps to know where it is and confirm you can reach it:

<https://github.com/kieDotson/tdwi-agentic-ai-workshop>

- Comfortable with Git? Clone it: `git clone https://github.com/kieDotson/tdwi-agentic-ai-workshop.git`
- Not using Git? On the repo page, click the green **Code** button, then **Download ZIP**.
- Either way, confirm you can see folders for all four labs. You don't need to open or run anything yet.

How the labs work

Each lab is **self-contained**. Every notebook opens with the finished state of the previous lab already loaded, so if you fall behind or a step runs long, it never blocks the next lab. You don't need to finish one lab to succeed at the next — come as you are, and pair up if you'd like. The first thing we do together is make sure everyone's environment works before we build anything.

You do **not** need an API key ahead of time. Keys are provided in the room on the day and removed afterward.

Installing Langflow Desktop

This is the full version of Step 1. It takes about five minutes. Doing it before class means we can start on time.

The version that matters is 1.11

Download the Desktop app using the button on the official page. That currently installs version 1.11.0, which is exactly what we want.

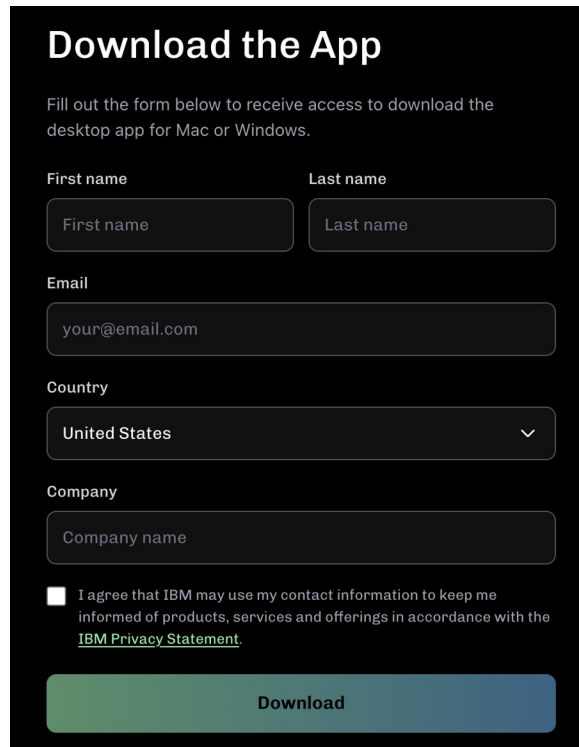
You don't need to hunt for a specific sub-version. Any 1.11.x is fine. If your installed app shows 1.11.0, you're correctly set up — keep it there and don't update.

A • Open the download page

Go to the official Langflow desktop download page:

<https://www.langflow.org/desktop>

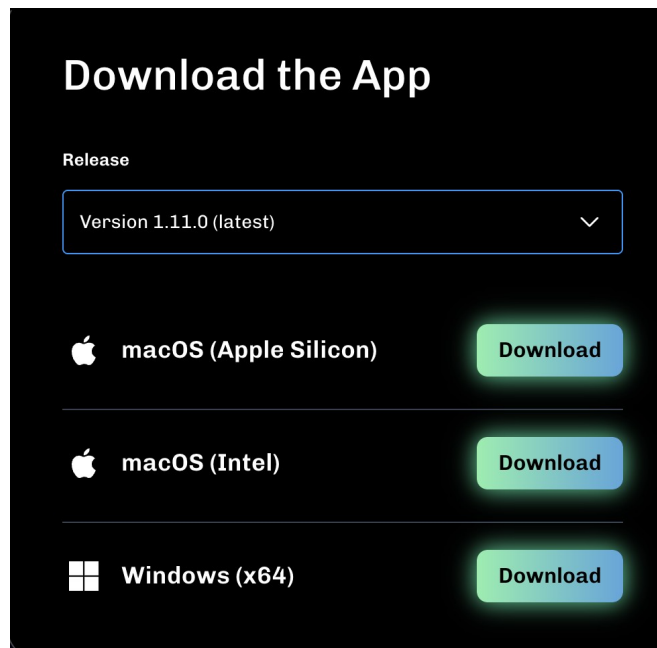
You'll first see a short form. Fill in your name, email, country, and company, accept the privacy checkbox, and click **Download**.

A screenshot of a dark-themed web form titled "Download the App". The form contains several input fields: "First name" and "Last name" (two separate boxes), "Email" (a single box with placeholder text "your@email.com"), "Country" (a dropdown menu showing "United States"), and "Company" (a single box with placeholder text "Company name"). Below these fields is a checkbox with the text "I agree that IBM may use my contact information to keep me informed of products, services and offerings in accordance with the IBM Privacy Statement." At the bottom of the form is a large, rounded rectangular button labeled "Download".

The access form you complete before the download links appear.

B • Choose your platform — leave the version at 1.11.0

After the form, a **Download the App** screen appears. At the top, the **Release** dropdown should already show **Version 1.11.0 (latest)**. Leave it on that.



Version 1.11.0 (latest) selected, with a Download button for each platform.

- **macOS (Apple Silicon)** — Macs from roughly 2021 onward (M1, M2, M3, M4).
- **macOS (Intel)** — older Intel-based Macs.
- **Windows (x64)** — most Windows 10 and 11 laptops.

Not sure which Mac you have?

Click the Apple menu in the top-left corner and choose "About This Mac." If the Chip line says "Apple M1/M2/M3/M4," choose Apple Silicon. If it says "Intel," choose Intel.

macOS: two requirements to know about

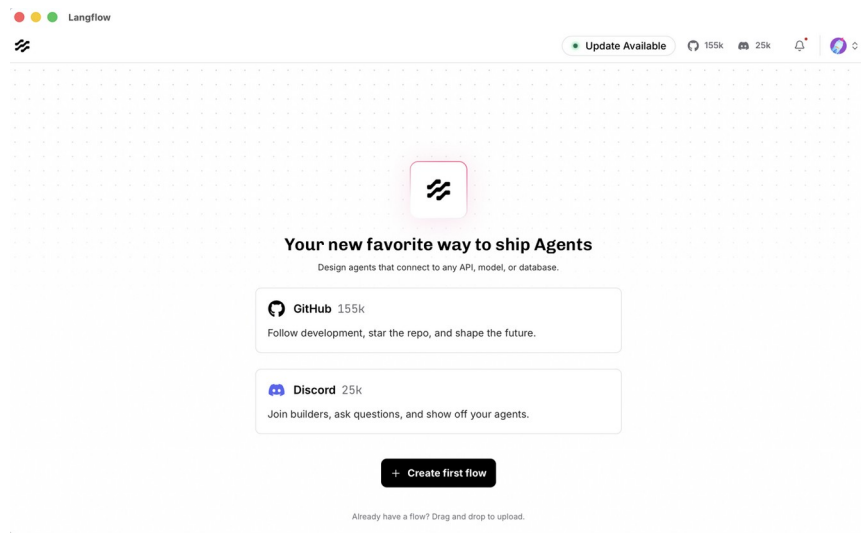
- Langflow Desktop requires macOS 13 (Ventura) or later.
- On first launch it may show "Xcode Command Line Tools required!" This is normal. Open Terminal, run `xcode-select --install`, follow the prompt, then open the Xcode app once so it finishes setup. Restart Langflow.

Windows: you may be asked for a C++ compiler

On first launch, Windows may show "C++ Build Tools Required!" Follow the on-screen prompt to install Microsoft C++ Build Tools, then reopen Langflow. This is a one-time step.

C · Open the app

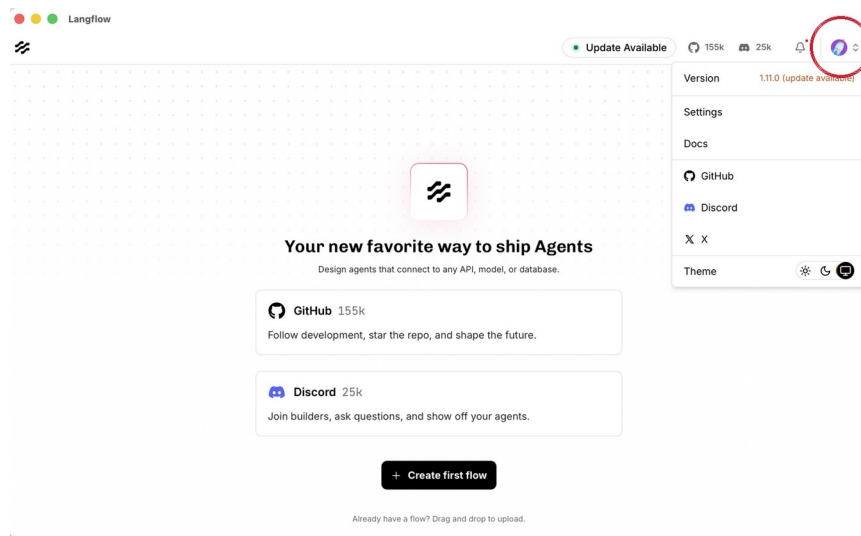
Launch Langflow after it installs. You'll land on a welcome screen with the heading "*Your new favorite way to ship Agents*" and a **Create first flow** button near the bottom. If you see this screen, the app installed correctly.



The Langflow home screen after a successful install.

D • Confirm the version is 1.11.0

This is the step that prevents problems in class. In the **top-right corner**, click the small rocket-ship icon (circled in red below), next to the up-down arrows. A menu opens with **Version** listed at the top. It should read **1.11.0**.



Click the rocket-ship icon (circled) to open the menu; Version shows 1.11.0 at the top.

If it says "update available," leave it alone

Langflow will offer to update to a newer version. Don't accept it. We're staying on 1.11.0 for the whole class so everyone's screens match and the lab file loads without errors. An "update available" note next to 1.11.0 is expected and fine.

You're ready for Lab 1 when:

- Langflow Desktop opens to the welcome screen.
- The rocket-ship menu shows Version **1.11.0**.
- You have **not** clicked update.

If something goes wrong

These are the error dialogs people run into most, and the fix for each:

- **"Xcode Command Line Tools required!"** (Mac) — run `xcode-select --install` in Terminal, open Xcode once, then restart Langflow.
- **"C++ Build Tools Required!"** (Windows) — follow the prompt to install Microsoft C++ Build Tools, then reopen Langflow.
- **"Python Environment Failed"** — make sure Python 3.10 or later is installed and available, then reinstall Langflow Desktop.
- **"Langflow Startup Failed"** — usually means port 7860 is already in use. Close any other Langflow instance. To check: `lsof -i :7860` on Mac, or `netstat -ano | findstr :7860` on Windows.

Still stuck? Don't spend the night on it — bring the laptop and we'll sort it in the first few minutes of class. But please do try the steps above first, since setup time is limited.

Advanced: other ways to install Langflow (optional)

Most people should stop at the Desktop app

The Desktop app is the recommended path and the one we support on the day. The two options below run Langflow as a local server from the command line instead. They're for people already comfortable with a terminal, Python, and virtual environments. You do not need them for the workshop.

Option A — Python package (pip via uv)

Requirements before you start:

- **Python 3.10-3.12 installed.** Avoid 3.13 — some dependencies fail to build on it.
- **uv** installed (Langflow's recommended package manager).
- On Mac, the same Xcode Command Line Tools noted above.

Then, from a fresh folder:

```
uv venv langflow-env
source langflow-env/bin/activate (Mac/Linux)
langflow-env\Scripts\activate (Windows)
uv pip install langflow==1.11.0
uv run langflow run
```

This runs a server, not an app window

There's no separate application to open. Once it starts (it can take a few minutes), go to `http://127.0.0.1:7860` in your browser — that's Langflow. This is the step people miss when coming from the Desktop instructions. If `langflow==1.11.0` won't resolve, the Desktop app is the reliable path — use that instead rather than troubleshooting versions.

Option B — Docker

If you already run Docker, start a container with:

```
docker run -p 7860:7860 langflowai/langflow:1.11.0
```

Then open **http://localhost:7860** in your browser. As with the pip path, this is a server — there's no separate app window.